In [2]:
```
import pandas as pd
import numpy as np
from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import silhouette_score
import matplotlib.pyplot as plt
import seaborn as sns
```

In [3]:
```
df = pd.read_excel("Rework_cummulative_brand_data.xlsx")
```

In [4]:
```
df.columns = df.columns.str.strip()
df.columns
```

Out[4]:
```
Index(['Brand', 'Quantity ( Grams)', 'MRP', 'Price', 'Price/100 Grams',
       'Discount (%)'],
      dtype='object')
```

In [5]:
```
features = ['Quantity ( Grams)','Price']
df_clean = df.dropna(subset=features).copy()
df_clean
df_clean.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 448 entries, 0 to 447
Data columns (total 6 columns):
 #   Column             Non-Null Count  Dtype  
---  ------             --------------  -----  
 0   Brand              448 non-null    object 
 1   Quantity ( Grams)  448 non-null    float64
 2   MRP                448 non-null    int64  
 3   Price              448 non-null    int64  
 4   Price/100 Grams    448 non-null    float64
 5   Discount (%)       448 non-null    int64  
dtypes: float64(2), int64(3), object(1)
memory usage: 21.1+ KB
```

In [6]:
```
X = df_clean[features]
X
```

Out[6]:

	Quantity (Grams)	Price
0	70.0	79
1	33.0	29
2	20.0	27
3	200.0	339
4	70.0	79
...	...	...
443	250.0	301
444	100.0	109
445	100.0	109
446	30.0	30
447	100.0	109

448 rows × 2 columns

In [7]:

```
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
X_scaled
```

```
Out[7]: array([[-5.73053746e-01, -3.62152440e-01],
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[-9.47458606e-02, -1.04720188e-01],
[ 1.00904157e+00,  1.26492112e-01],
[-7.57018318e-01, -4.12208712e-01],
[-9.47458606e-02, -2.96357813e-02],
[ 1.00904157e+00, -1.07103820e-01],
[-7.01828946e-01, -3.81221496e-01],
[-9.47458606e-02, -1.47625564e-01],
[-9.47458606e-02, -1.83380043e-01],
[ 8.92187108e-02, -2.64423530e-01],
[-9.47458606e-02, -1.36899220e-01],
[-6.46639575e-01, -4.01482368e-01],
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[ 1.74489985e+00, -1.59543723e-01],
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[-6.83432489e-01, -3.70495152e-01],
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[-4.62675004e-01, -3.26397961e-01],
[-7.20225404e-01, -4.20551423e-01],
[-4.62675004e-01, -3.26397961e-01]])
```

In [8]:
```
wcss = []
k_range = range(2,11)  # Initialize BEFORE the loop

for k in k_range:
    K_means = KMeans(n_clusters=k)
    K_means.fit(X_scaled)
    wcss.append(K_means.inertia_)

plt.subplot(1, 2, 1)
plt.plot(k_range, wcss, marker="o", linestyle="--")
plt.title("Elbow Bend Method")  # also fix the typo: plt.tite → plt.title
plt.xlabel("Clusters")
plt.ylabel("WCSS")
plt.show()
```

```
c:\ProgramData\anaconda3\Lib\site-packages\sklearn\cluster\_kmeans.py:1419: UserWarn
ing: KMeans is known to have a memory leak on Windows with MKL, when there are less
chunks than available threads. You can avoid it by setting the environment variable
OMP_NUM_THREADS=2.
  warnings.warn(
c:\ProgramData\anaconda3\Lib\site-packages\sklearn\cluster\_kmeans.py:1419: UserWarn
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chunks than available threads. You can avoid it by setting the environment variable
OMP_NUM_THREADS=2.
  warnings.warn(
```

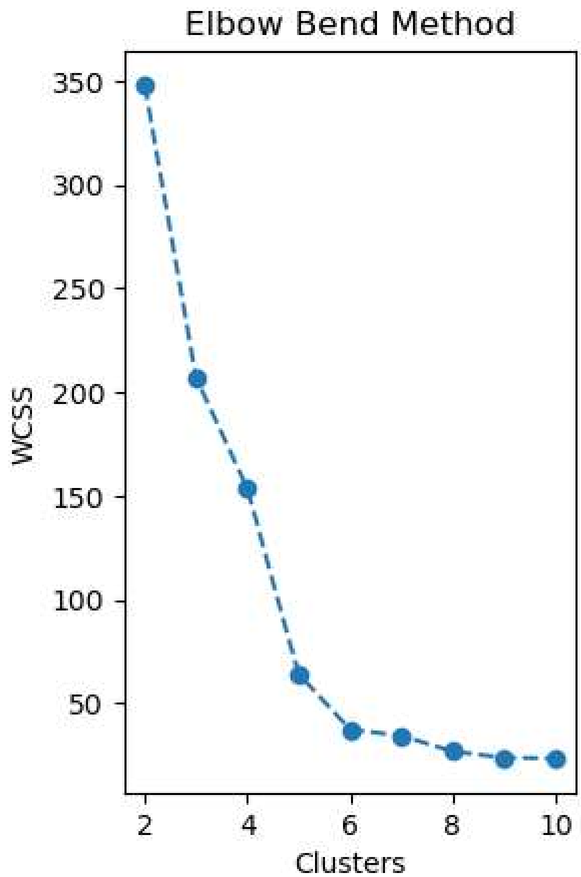

In [9]:
```
sil_scores = []
k_range = range(2,11)
for K in k_range :
    k_means = KMeans(n_clusters=K,random_state=42)
    labels = k_means.fit_predict(X_scaled)
    sil_scores.append(silhouette_score(X_scaled,labels))
plt.plot(k_range,sil_scores,marker="o",linestyle="--")
plt.title("silhouette_score")
plt.xlabel("Clusters")
plt.ylabel("silhouette_scores")
plt.grid(True)
plt.show()
```

```
c:\ProgramData\anaconda3\Lib\site-packages\sklearn\cluster\_kmeans.py:1419: UserWarn
ing: KMeans is known to have a memory leak on Windows with MKL, when there are less
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c:\ProgramData\anaconda3\Lib\site-packages\sklearn\cluster\_kmeans.py:1419: UserWarn
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OMP_NUM_THREADS=2.
  warnings.warn(
```

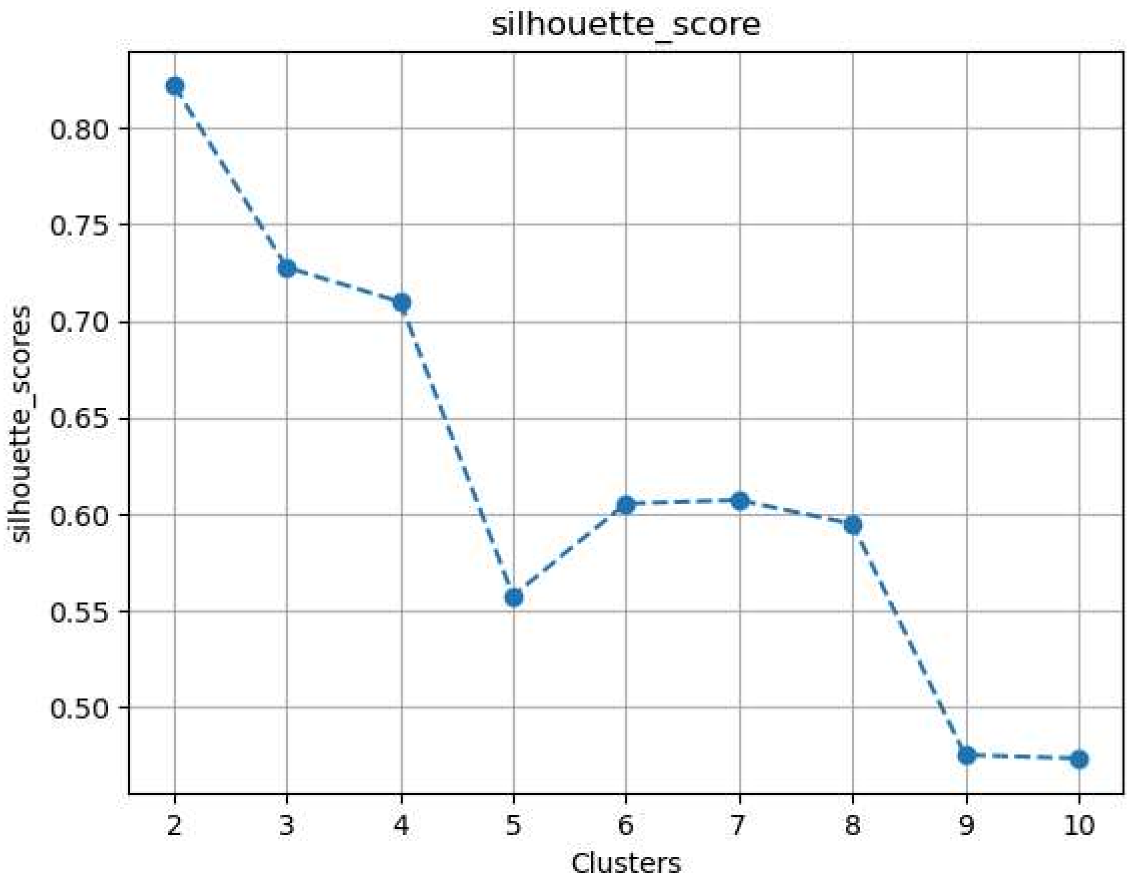

In [12]:
```
final_cluster = KMeans(n_clusters=4,random_state=42)
clusters = final_cluster.fit_predict(X)
df["Cluster"] = clusters
```

```
c:\ProgramData\anaconda3\Lib\site-packages\sklearn\cluster\_kmeans.py:1419: UserWarn
ing: KMeans is known to have a memory leak on Windows with MKL, when there are less
chunks than available threads. You can avoid it by setting the environment variable
OMP_NUM_THREADS=2.
  warnings.warn(
```

In [14]:
```
df
df.to_excel("Rework_cummulative_brand_data_cluster.xlsx")
```